Core Base: Why Bioinformatics Requires Multiple Disciplines?

Bioinformatics is interdisciplinary because biological problems are too complex for any single field to solve alone.

- **Biology** tells us what questions to ask and helps interpret results. A biologist understands that a DNA sequence isn't random but it has functional meaning. They know which genes matter for disease and can recognize when computational predictions make biological sense.
- **Computer science** provides the tools to handle massive datasets. A single genome has billions of bases impossible to analyze manually. Computer scientists develop algorithms like BLAST that search through millions of sequences in seconds. They build databases to store and retrieve biological data efficiently.
- **Mathematics and statistics** help us distinguish real discoveries from random chance. When BLAST returns an E-value of 1e-50, statistics tells us this match is practically impossible to occur by accident. Statistical methods also help analyze gene expression data and identify meaningful patterns.

For example, the Human Genome Project required biologists to understand genetics, computer scientists to assemble DNA fragments, mathematicians to handle sequencing errors, and statisticians to validate results. Each discipline contributed essential expertise that others couldn't provide.

Question 1: DNA Sequence Associated with Rare Genetic Disorder

Recommended Databases: NCBI GenBank, OMIM, ClinVar

The researcher should start with **NCBI BLAST** against GenBank. This database contains all publicly available DNA sequences. BLAST will quickly tell if this sequence has been reported before. If matches exist, the researcher can see which gene the sequence belongs to and what organisms carry it.

Next, the researcher should check **OMIM (Online Mendelian Inheritance in Man)**. This database specifically catalogs human genes and genetic disorders. If the sequence is linked to a disease, OMIM provides detailed clinical descriptions, inheritance patterns, and references to scientific literature. For rare disorders, OMIM is particularly valuable because it collects information that might not appear elsewhere.

ClinVar is also essential because it records how specific sequence variations affect health. It tells whether a variation is "pathogenic," "benign," or "uncertain significance." This helps the researcher understand if their sequence variation is likely causing the disorder.

Using these three databases together gives a complete picture; whether the sequence is known, what gene it's in, what disease it causes, and how variations in it affect patients.

Question 2: Comparing Bacterial Genomes for Conserved Genes

Recommended Databases: NCBI GenBank/RefSeq, KEGG

The scientist should use **NCBI GenBank** to download complete genome sequences of related bacterial species. GenBank provides fully annotated genomes with gene locations and predicted functions. RefSeq is particularly reliable because its sequences are curated and non-redundant.

KEGG (Kyoto Encyclopedia of Genes and Genomes) is the most appropriate database for identifying conserved genes. KEGG uses KO (KEGG Orthology) numbers; each KO represents a gene with a specific function that has been conserved across species. If multiple bacteria share the same KO, they likely share the same functional gene.

KEGG also maps genes onto biological pathways. This means the scientist can see not just which genes are conserved, but what those genes do together. For example, they might find that all related bacteria share KOs for energy metabolism, but only the disease-causing species has KOs for toxin production. This reveals which genes are essential for survival (the conserved ones) and which are specific to the disease.

NCBI provides the raw genomic data, while KEGG helps interpret the biological meaning of conserved genes. Together, they allow the scientist to distinguish core genes (present in all species) from accessory genes (unique to specific strains).

Question 3: Protein from a New Microorganism

Recommended Databases: UniProtKB, NCBI Protein BLAST, InterPro

The company should first use **UniProt BLAST** to search against UniProtKB, the largest protein database. UniProt has two parts: Swiss-Prot contains manually reviewed proteins with reliable annotations, while TrEMBL contains automatically annotated proteins from genome projects.

If the protein matches a Swiss-Prot entry, the company can get detailed information: protein name, function description, domain structure, subcellular location, and references to published research. For example, if the protein is 80% identical to a known enzyme, the company can confidently assume their protein has similar properties.

If no close matches exist, **InterPro** helps identify conserved domains within the protein. Even if the whole protein is novel, it might contain functional building blocks, like a kinase domain or DNA-binding domain, that reveal what it does.

NCBI Protein BLAST provides additional coverage beyond UniProt and may find matches in sequences that haven't made it into UniProt yet.

The company will learn whether their protein is truly novel or belongs to a known family. If it's novel, they've potentially discovered a new enzyme with unique properties valuable for biotechnology applications. If it's known, they can leverage existing knowledge to characterize it faster.

Question 4: Determining Function of Unknown Protein

Recommended Databases: UniProt, NCBI BLAST, KEGG, InterPro

The scientist should start with **NCBI BLAST** against UniProtKB. If the protein matches a known protein with high identity, the function is already determined. the scientist simply reads the annotation in UniProt. UniProt entries provide function descriptions, enzyme classifications, interaction partners, and literature references.

If BLAST matches are weak, the scientist uses **InterPro** to identify conserved domains. For example, finding a zinc finger domain suggests DNA binding; finding a kinase domain suggests phosphorylation activity. Combining domain information helps infer function even without a close overall match.

KEGG provides pathway context. If the protein is assigned a KO number, KEGG shows which biological pathways it participates in. This tells the scientist not just what the protein does, but where it fits in the cell's larger systems.

For example, a protein might match a known kinase at 40% identity. InterPro confirms it has a kinase domain. KEGG shows it maps to a signaling pathway. Together, these clues suggest the protein is a signaling kinase. The scientist can then design experiments to test this hypothesis.

The combined information from these databases builds a complete functional picture from molecular activity to biological role.

Question 5: Identifying an Unknown Biological Sequence

Step-by-Step Approach:

Step 1: Determine Sequence Type
First, check the letters in the sequence. DNA contains A, T, G, C only. RNA contains A, U, G, C. Protein contains 20 different amino acid letters. If uncertain, run both nucleotide and protein BLAST searches to see which produces meaningful results.

Step 2: Search Against General Databases
For DNA or RNA, use **NCBI BLAST (blastn)** against the nucleotide database. This will find identical or similar sequences. For protein, use **UniProt BLAST (blastp)** . If the sequence is

DNA but might code for protein, use **blastx** (translates DNA to protein and searches protein databases). This catches similarities that nucleotide BLAST might miss due to codon differences.

Step 3: Identify the Gene or Protein
Based on BLAST results, find the gene name or protein name. Use the **NCBI Gene database** for DNA sequences or **UniProt** for protein sequences to get detailed annotation. If the sequence is DNA and contains an open reading frame, it likely encodes a protein. If no ORF exists, it might be regulatory DNA or non-coding RNA.

Step 4: Determine Organism and Function
Use BLAST results to identify the organism of origin or closest relative. Check the **NCBI Taxonomy database** for evolutionary relationships. For function, use **KEGG** to see pathway involvement or **InterPro** for domain analysis.

This stepwise approach works because each stage builds on the previous. Starting with sequence type prevents wasting time searching the wrong database. Using BLAST first quickly identifies matches if they exist. Domain and pathway analyses provide deeper understanding when simple matches aren't enough.

For example, a sequence might initially look like DNA. BLAST finds no nucleotide matches. But blastx finds a match to a bacterial protein, revealing it's actually a DNA sequence encoding a novel protein. InterPro then identifies its domains, and KEGG shows its pathway. Within an hour, the unknown sequence becomes a characterized bacterial enzyme without any lab experiments.